

# DIPESH KUNWAR

dipeshkunwar8@gmail.com | (+91) 966-4502-972 | github.com/Dipesh038 | linkedin.com/in/dipesh-kunwar-a09894322

## PROFESSIONAL SUMMARY

Motivated Computer Science student with a strong foundation in Core Java, Object-Oriented Programming (OOP), and backend software development. Proficient in designing and implementing database-driven applications using JDBC, MySQL, and Java Collections Framework with emphasis on clean architecture and Data Structures & Algorithms. Skilled in problem-solving, debugging, and writing maintainable Java code; seeking a Java Backend Developer or Java Developer Intern role to contribute to real-world software development projects.

## TECHNICAL SKILLS

**Programming Languages:** Java, Core Java, SQL

**Backend Technologies:** JDBC, Java Collections Framework, Backend Development, Software Development

**Databases:** MySQL, Relational Database Design, CRUD Operations

**Concepts:** OOP, Data Structures & Algorithms, Problem Solving, Debugging

**Tools:** Git, GitHub, Visual Studio Code, Android Studio

## PROJECTS

### Student Management System | Core Java | JDBC | MySQL | OOP

- Designed and developed a full-stack console-based backend application to manage student enrollment, grade tracking, and record search, demonstrating end-to-end software development using Java.
- Implemented CRUD operations via JDBC with a normalized, relational MySQL schema, applying database design best practices to ensure data integrity and optimized SQL query performance.
- Applied OOP principles — encapsulation, inheritance, and polymorphism — to architect a modular, multi-layered Java codebase with clear separation of business logic, data access, and presentation.

### Library Management System | Core Java | Java Collections Framework | DSA

- Developed a console-based library backend system supporting book cataloguing, member registration, and issue/return tracking, utilizing core Java and advanced data structure design.
- Leveraged Java Collections Framework (LinkedList, TreeMap, HashSet) to efficiently manage large datasets, achieving  $O(\log n)$  search and sort performance through algorithmic optimization.
- Implemented custom exception handling and input validation modules to ensure robust, crash-resistant program execution, reinforcing strong debugging and problem-solving skills.

### Inventory Management System | Core Java | JDBC | MySQL | Backend Development

- Developed a menu-driven inventory backend system to manage product stock, supplier records, and transaction logs, showcasing practical Java backend development and software engineering skills.
- Integrated JDBC for persistent MySQL storage using parameterized queries to prevent SQL injection attacks, demonstrating security-aware backend development and database design proficiency.
- Designed a normalized relational database schema with foreign key constraints and indexed tables, reflecting solid understanding of DBMS principles and optimized query execution.

## EDUCATION

### Pandit Deendayal Energy University

July 2023 – May 2027

B.Tech – Computer Science and Engineering | CGPA: 7.77 / 10.00

Relevant Coursework: Object-Oriented Programming (OOP), Database Management Systems (DBMS), Data Structures & Algorithms (DSA), Operating Systems, Discrete Mathematics

## CODING PROFILES

**LeetCode:** [leetcode.com/Dipesh038](https://leetcode.com/Dipesh038) — 70+ Problems Solved (Arrays, Strings, LinkedList, OOP)

**HackerRank:** [hackerrank.com/Dipesh038](https://hackerrank.com/Dipesh038) — Java (Basic) Certification

**GitHub:** [github.com/Dipesh038](https://github.com/Dipesh038) — 3 Backend Projects (Java, JDBC, MySQL)

## ACHIEVEMENTS & CERTIFICATIONS

- Data Structures and Algorithms (Java):** Completed dedicated certification focused on algorithmic problem-solving, computational complexity, and DSA implementation in Java.
- Rural Internship – Future Technology:** Applied emerging software and technology solutions to real-world challenges in underserved communities; recognized for practical problem-solving impact.
- Android Development Certification:** Gained foundational knowledge of mobile application architecture, Java-based Android development, and software design patterns.
- Healthcare Technology Certification:** Completed training on the application of software technologies within the healthcare domain, broadening understanding of backend system requirements.